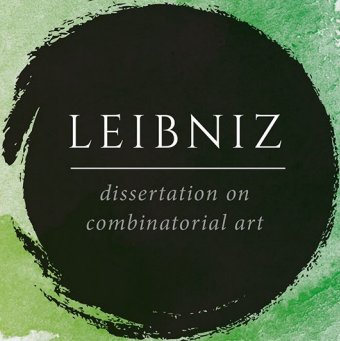


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LEIBNIZ

*dissertation on
combinatorial art*

TRANSLATED WITH INTRODUCTION AND COMMENTARY BY

Massimo Mugnai, Han van Ruler,
and Martin Wilson

Leibniz: Dissertation on Combinatorial Art

LEIBNIZ FROM OXFORD

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Translated with introduction and commentary by

MASSIMO MUGNAI, HAN VAN RULER,
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Preface

History is bound to cherish the youthful work of a later genius, especially if its themes anticipate those of masterpieces to come. In the case of Leibniz's *Dissertatio de arte combinatoria* (1666), however, scholarship has been curiously reserved in dealing with this particular *travail de jeunesse*. Although studies by such eminent Leibniz specialists as Louis Couturat (1868–1914), Karl Dürr (1875–1928), and Eberhard Knobloch have added much to the interpretation of Leibniz's first publication in print, a critical edition of the *Dissertatio* has thus far been wanting, no doubt due to its arduous phrasing, its wide-ranging subject matter, as well as its complex overall theme.

All of these factors make Leibniz's *De arte combinatoria* (henceforth *DAC*) a text that is not easy to edit. Working on the present edition, moreover, the editors came across many further obscurities complicating an annotated rendition of the work. Its mix of languages and subject matters is only a minor hurdle in comparison to the strenuous way in which Leibniz at times gives expression to the ideas contained in it, which, for instance, may include descriptions in Latin of complex mathematical operations. Another problem encountered was to find the many texts by authors cited in the *DAC*, a difficulty that was enhanced by Leibniz's recurrent habit of quoting authors indirectly—and on occasion incorrectly.

The wealth of materials in *De arte combinatoria* and the dense argumentation it contains did not impede Martin Wilson to work on an English translation of the text over a period of many years, until he finally suggested to prepare it for publication in the wake of the appearance of Arnold Geulincx's *Ethics* in 2006, a project on which he and Han van Ruler had been working together.¹ Finding numerous puzzling details in Leibniz's text and a multitude of references unaccounted for in the extant secondary literature, it would take Martin Wilson and Han van Ruler some further years of textual and bibliographical investigations to come up with a reasonably reliable translation. At this point, Massimo Mugnai joined Martin

¹ Geulincx (2006).

and Han and the project began to take the shape of an edition aiming not only to satisfy the requirements of the inner circle of Leibniz's scholars but a larger spectrum of readers interested in Leibniz's philosophy and the origins of the combinatorial calculus in general. This implied not only the obvious task of preparing an intelligible translation of the *DAC*, but even that of explaining the many difficult passages of the text. As a corollary to this task, it became indispensable to find all writings and authors mentioned by Leibniz, some of them very obscure or almost unknown, and to equip the text with a rich apparatus of footnotes.

Thus, the annotated translation that lies before you is the product of a collective enterprise in which Martin Wilson has been the principal translator of Leibniz's text. Massimo Mugnai and Han van Ruler have revised and edited the translation to produce the text as it now stands, and jointly contributed to its annotation. Han has taken care of the largest part of the footnotes, whereas Massimo, besides preparing the Latin version of the *DAC* here included, has provided the main text with an Introduction, aiming to situate in its historical context this early work of Leibniz and to present the main results contained in it.

In the process of the collaborative work on the project, Martin Wilson was sadly found missing for a period, due to a stroke he suffered on 1 March, 2011. Though quickly regaining most of his colourful personality, Martin has not been able to fully recover. The editors would like to express their gratitude to Astley Nursing Home in Worcestershire, England, for the care they have given Martin ever since, as well as to Barry Sweeting, who has been acting as Martin's representative and became a good friend.

The editors are also grateful to Matthias Armgardt, Albrecht Heeffer, Franco Montanari, Jan Papy, Horst Pfefferl, Wolfgang Schlosser, and Andrew Weeks for their helpful comments on some questions of detail. Richard Arthur, Vincenzo De Risi, Enrico Giusti, Enrico Pasini, and Monica Ugaglia have read a first draft of the Introduction, finding mistakes and kindly suggesting amendments. A special word of thanks is due to Mariano Giaquinta and Eberhard Knobloch for their generous advice on many technical questions.

Working on the annotation of the *DAC* during years in which there has been a stupendous increase in the number of electronic sources available online, Han van Ruler was nonetheless pleased to be helped by the cooperative staff of a huge number of libraries still important for their treasures—Leiden University Library, the Herzog August Bibliothek Wolfenbüttel, the British Library, and the Bibliothèque Nationale de France in particular.

Besides expressing his debt to the staff of these and many other libraries, he also wishes to thank Atsuko Fukuoka, Iqbal Faridi, and Peter Guttner, as well as Patrick Masereel and Vincent de Keijzer for their hospitality in Wolfenbüttel, London, and Paris over the years.

Finally, the editors are grateful to Maria Rosa Antognazza and Paolo Mancosu for their constant encouragement and to Peter Momtchiloff of Oxford University Press, whose confidence and dedication have given an enormous spurt to the realization of this long-term project.

Massimo Mugnai, Han van Ruler
Firenze and Leiden

On this Edition

The Latin text here presented of Leibniz's *Dissertatio de arte combinatoria* is faithful to the original publication of the DAC (Leipzig: Fick and Seubold, 1666), the full title of which reads:

Dissertatio De Arte Combinatoria, In qua Ex Arithmeticae fundamentis Complicationum ac Transpositionum Doctrina novis praeceptis extruitur, & usus ambarum per universum scientiarum orbem ostenditur; nova etiam Artis Meditandi, Seu Logicae Inventionis semina sparguntur. Praefixa est Synopsis totius Tractatus, & additamenti loco Demonstratio Existentiae Dei ad Mathematicam certitudinem exacta.

In a new, ameliorated edition, the text of the DAC was reproduced by Carl Immanuel Gerhardt (1816–99) in volume IV of his edition of *Die philosophische Schriften von Gottfried Wilhelm Leibniz* (Berlin: Weidmann, 1880 / Hildesheim: Georg Olms, 1960), 27–104.

In the text below, however, the reader will find references in square brackets (in bold) to the 1666 edition and also square brackets (in italics) to the page numbers of the *Akademie*-edition of the DAC:

Gottfried Wilhelm Leibniz, *Sämtliche Schriften und Briefe* herausgegeben von der Deutschen Akademie der Wissenschaften zu Berlin, Sechste Reihe, Erster Band, Darmstadt: Reichl, 1930 / Berlin Akademie Verlag, 163–230.

In the translation here presented, Leibniz's typography has been followed almost literally, even if the structure of individual sentences has been adapted in such ways as to yield a readable English text.

In dealing with typos, omissions, miscalculations, and mistaken references in the 1666 original, as well as with Leibniz's recurrent inclusion of Greek and German words into the Latin text, consistency in referring to these has not been an aim in itself. Obvious editorial interventions will be marked by the use of square brackets in the text of the translation.

In choosing case by case whether simply to copy the original or to make editorial interventions, it has been the editors' intention to offer an intuitively

understandable account of what the author must have had in mind. Latin text, translation, and annotations will hopefully provide the reader with sufficient materials to do the same and reconstruct the mathematical and philosophical objectives Leibniz set himself when writing the *DAC*.

As far as the editors know, besides the present edition there is only one entire translation of the *DAC* into a modern language: G. W. Leibniz, *Dissertación de arte combinatoria*, translated by Manuel A. Correia, in G. W. Leibniz, *Obras filosóficas y científicas. Escritos Matemáticos*, (ed.) Mary Sol de Mora Charles (Granada: Editorial Comares), 2015, vol. 7B, 547–643. A previous translation into Catalan was edited, again by Manuel Correia, in 1992.

Quotation marks

In what follows, double quotation marks have generally been used for quotations, and single quotation marks for highlighting (the use of) specific concepts or expressions. Double quotation marks, however, may occasionally also point out odd or unusual expressions.

In the footnotes, double quotation marks are also used for quotations in Latin, but quotations of Latin verse and references to (the use of) Latin concepts and expression are generally given in *italics*.

Finally, Leibniz himself uses double quotation marks to indicate the solutions he offers to the problems dealt with in the *DAC*—a custom that the editors have chosen to copy in the translation.

References

Next to page number references (after ‘:’) and references to volumes (‘vol.’, ‘vols’), the reader will also find references to the signature marks (‘sig.’, ‘sigg.’), which are found especially in the front matter of early-modern prints, as well as, where applicable, to columns rather than pages (‘col.’, ‘cols’).

Abbreviations

A list of bibliographical abbreviations and abbreviated titles for reference works occurring in the notes to both the Introduction and the Translation of the *DAC*, may be found in sections A. (*Leibniz Works*) and D. (*Abbreviations*) of the Bibliography, pp. 287 and 301–2, below.

Contents

Introduction	1
A Dissertation on Combinatorial Art	58
Text and Translation	62
<i>Bibliography</i>	287
<i>Index</i>	303

Introduction

1 The project for the ‘characteristic art’

Leibniz was twenty years old when the *Dissertation on Combinatorial Art* (henceforth *DAC*) was first published. As he would later write: “the booklet was composed when the author was still young, in the year 1665, and was edited at Leipzig in the year 1666”.¹ The work expanded the text of a dissertation he had defended earlier that same year at the Leipzig Faculty of Philosophy, aiming for a university position. This first embryonic form of the *Dissertation* was entitled *Disputatio Arithmetica de complexionibus* (‘Arithmetical Disputation on Complexions’) and contained the first part of the 1666 work, from *Cum Deo* (“With God’s Help”) through to the end of section 8 of the first problem.²

At the time, Leibniz was not particularly well trained in mathematics, as he was to admit later on the occasion of a re-edition of the *DAC*. When, indeed, the *DAC* was published again without the author’s permission by the Frankfurt book dealer Cröker in 1690, this provoked a reaction on Leibniz’s part, who wrote a short notice in the *Acta Eruditorum*.³ As he writes there, the booklet was “not sufficiently polished” and no longer agreed with the ideas he held “at present” (1691). Moreover, the book left something to be desired:

...some problems concerning numbers could have been solved in a far better way and the solutions are susceptible of more accurate proofs. At the time, indeed, [the author] had only a vague idea of higher mathematics and thus, unaware of the discoveries of others and not sufficiently acquainted

¹ GP 4: 103. The original (see below, note 4) has ‘1668’.

² Cf. A VI, 1: 170–5. The *Arithmetical Disputation* included four more corollaries (A VI, 1: 228–30). The entire title of the *Arithmetical Disputation* was: *Disputatio Arithmetica de Complexionibus, Quam in Illustri Academia Lipsiensi Indultu Amplissimae Facultatis Philosophicae pro loco in ea obtinendo prima vice habebit* M. Gotifredus Guilielmus Leibnuzius Lipsiensis, J. U. Baccal. D. 7. Martii Anno 1666. H. L. Q. C. Lipsiae, Literis Sporelianis.

³ Cf. Knobloch (1974: 412).

with mathematical analysis, he was able to produce something of his own on the basis of a hasty reflection (...) ⁴

Despite this, however, Leibniz also notes that “the booklet” contains “many new meditations” that he does not regret, concerning the “art of discovery” (*ars inveniendi*) and the “excellent” idea of an alphabet of human thoughts.

The ‘art of discovery’ and the *alphabet of human thoughts* are two issues strongly connected with one another. They both pertain to the discipline of logic, for which Leibniz manifested a genuine interest from an early age onwards. According to an autobiographical passage included in a text probably written in the year 1682, Leibniz, when he was “not yet twelve”, was filling “sheets of paper with wonderful meditations about logic, attempting to overcome the subtleties of the scholastics.” ⁵ In the same vein, in a text composed during the years 1683–5, he writes:

As a boy I learned logic, and having already developed the habit of digging more deeply into the reasons for what I was taught, I raised the following question with my teachers. Seeing that there are categories for the simple terms by which concepts are ordered, why should there not also be categories for complex terms, by which truths may be ordered? I was then unaware that geometricians do this very thing when they demonstrate and order propositions according to their dependence upon each other. It seemed to me however that this could be achieved universally if we first had the true categories for simple terms and if, to obtain these, we set up something new in the nature of an alphabet of thoughts, or a catalogue of the highest genera or of those we assume to be highest, such as *a, b, c, d, e, f*, out of whose combination inferior concepts may be formed. ⁶

Later on, the idea of building an ‘alphabet of thoughts’ evolved into a very complex project, which remained a central topic among Leibniz’s philosophical interests until the end of his life: the composition of a ‘universal characteristic’, or ‘characteristic art’ (*ars characteristica*).

The Latin word *character* has the same meaning as the English words ‘mark’, ‘sign’; and with the expression *ars characteristica*, Leibniz refers to

⁴ Leibniz’s notice was published in the *Acta Eruditorum* (Leipzig) of the year 1691 (pp. 63–4, cf. GP 4: 103–4) and has been recently republished in Leibniz, *Essais*, vol. III: 995–6.

⁵ GP 7: 126. For “the age of twelve”, see what G. H. Parkinson writes in *Logical Papers*, p. x, n. 4, where, relying on Fischer-Kabitz (1920: 712), he considers it an exaggeration on Leibniz’s part.

⁶ L: 229 (A VI, 4: 538).

a system of symbols denoting concepts and propositions, on which one may operate according to certain rules to discover new truths and to find proofs for the old ones. To construct the *characteristic*, two fundamental steps were needed: 1) performing an analysis of all human concepts, so that we may have a complete list of all simple concepts; 2) recombining the simple concepts, thus generating all possible complex concepts and all possible truths. Step (1) requires that we dispose of a general inventory of all known concepts and truths, and this motivated Leibniz to plan a further step: 3) the construction of a general encyclopedia of human knowledge. Leibniz's optimistic project is that of analysing each complex concept in its component parts, until we reach the simple, primitive ones. At the first stage of the project, clearly witnessed by the *DAC*, Leibniz takes for granted (in accordance with the Lullist tradition) that we human beings could attain the first, simple concepts corresponding to the properties of God. Later on, however, around 1686, Leibniz states that, even though we cannot reach the simple concepts, we may stop our analysis at a certain point and consider as simple, i.e. primitive, those concepts that we cannot further analyse. Thus, in this case the simple concepts are simple *for us*, not absolutely.⁷

Once the encyclopedia were constructed, however, the problem would still remain of finding the rules to perform the analysis and re-combination of the concepts; and it is for this reason that Leibniz adds another stage to the previous ones: 4) the constitution of the so-called 'general science' (*scientia generalis*).

Leibniz thinks of the general science in terms of a very abstract discipline, i.e. as a kind of 'meta-science', which was to investigate the various methods and procedures (deductive and inductive, logical and empirical) internal to each scientific field and displayed by people involved in their respective research activities. Of course, the study of logic would play a pre-eminent role in this general science and it is therefore no wonder that Leibniz, in the period from 1676 to 1690, developed a series of essays on the logical calculus. These essays were aimed at finding the general rules and principles governing any kind of logical inference, and may be considered as the first successful attempt, in Western culture, to develop an algebraic treatment of logic before George Boole's seminal work.⁸

⁷ Cf., for instance, A VI, 4A: 742, 746, 964.

⁸ Boole (1847). On Leibniz's logical writings and Boolean algebras, see Castañeda (1990); Swoyer (1994); Lenzen (2004); Mugnai (2010); Malink-Vasudevan (2016), and the debate between Mugnai and Malink-Vasudevan (2017).

Almost in the same period as his inquiries on logical calculus, Leibniz applied himself to the task of determining the main features of the so-called ‘rational grammar’ (*grammatica rationalis*), which, according to his intentions, should constitute a fundamental tool toward the construction of a ‘universal language’.

In some texts, Leibniz seems to identify ‘universal language’ and ‘characteristic art’: he models the *characteristic* on the example of a natural language based on a grammar reduced to its essential elements. In general, however, Leibniz thinks of the construction of a universal language as a step toward the constitution of the universal characteristic, which, as we have seen, presupposes a large-scale program with lots of elements besides the universal language, including the encyclopedia, the general science, the logical calculus, etc.

If we compare what Leibniz writes in the *DAC* with the project for the characteristic art as it emerges in the years following his stay in Paris (March 1672–October 1676), it is not difficult to find in the 1666 work many seeds of the future developments. From the *DAC*, indeed, the project for the characteristic art inherits the following, fundamental ideas:

- 1) An analysis of concepts is needed to reach the first, simple concepts;
- 2) By means of a synthetic procedure, complementary to that of analysis, all possible complex concepts may be derived out of the simple ones;
- 3) The process of synthesis can be performed by combining the concepts, simple and complex, on the basis of the rules of logic;
- 4) If we assign to each simple concept a symbol of an alphabet, we may unambiguously represent in symbolic form any complex concept whatsoever;
- 5) Given that any elementary truth can be expressed combining two concepts corresponding, respectively, to a subject and a predicate, the process of combining the simple concepts will generate all possible truths, constituting the basis of the *art of discovery*;
- 6) If we employ *numbers* to represent concepts, we may make logical inferences simply operating on numbers, as in an arithmetical calculus, instead of operating directly on concepts.

To appreciate the novelty and originality of Leibniz’s project, as it is articulated in the *DAC*, however, we have to dwell a little on the cultural background against which these ideas were conceived.

2 'Lullism' in the seventeenth century: art of memory and first steps towards the creation of an encyclopedia

In the first half of the fifteenth century, humanists strongly reacted against scholasticism in general and to scholastic logic in particular, which they tended to consider as a discipline emblematic of a philosophical attitude that was preoccupied primarily with frivolous topics and exclusively devoted to produce sophistries. With the rediscovery of the works of Euclid and the publication of texts by authors of late antiquity such as Heron of Alexandria and Pappus, the standard of rigorous reasoning ceased to be identified with Aristotle's *Organon*.

This does not mean, however, that, during the middle ages, Euclid's *Elements* were unknown. In the first half of the twelfth century Adelard of Bath translated into Latin an Arabic manuscript of the *Elements*. Adelard's edition was followed by several other translations made from Arabic sources and by the translation made by Gerardo da Cremona. In the thirteenth century, Campano da Novara prepared a text of Euclid's work, which was later employed by Luca Pacioli and Nicolò Tartaglia for their editions. Medieval commentaries on the *Elements*, however, with the exception of those belonging to the Arabic tradition, were usually quite poor. In the second half of the sixteenth century, the *Elements* were translated into modern languages.⁹ In the same period, Greek manuscripts began to circulate in Europe and Bartolomeo Zamberti translated the *Elements* into Latin directly from the Greek (1505), thus opening a new era in the philological reception of the text. In 1533 Simon Grynaeus published the *editio princeps* of the Greek text. In this sense it seems proper to speak of a 'rediscovery' of Euclid in the Western World.¹⁰

Books on logic from the fifteenth and sixteenth century often expressed the need to develop old-fashioned logical matters according to the method followed by mathematicians. In the introduction to his treatise on logic (first published in 1492), Girolamo Savonarola (1452–98), for example, criticizes the obscurity that characterized the traditional way of teaching logic and states that he aims to offer a synthesis of the entire dialectic 'according to the usual practice of mathematicians'.¹¹ Thus, it is no wonder that Jacob

⁹ Into Italian (1543), German (1562), French (1564), English (1570), Spanish (1576), and Dutch (1606).

¹⁰ See De Risi (2016).

¹¹ Savonarola (1982: 3) complains that many people abstain from studying logic because of the intrinsic difficulties of the discipline, "the obscurity of Aristotle's books and the sophistries

Pelletier (1515–82), more than fifty years after Savonarola's logic, in his introduction to the Latin translation of the *Elements* (1557),¹² writes that “every proof which leads us to truth is geometrical in character”, and that “we are not able to distinguish the true from the false if we have not previously been well acquainted with Euclid.”¹³ From the beginning of the sixteenth century onwards, even though in many cases logic continued to be taught along traditional lines, authors on the subject of logic start raising the issue of its relationships with mathematics.

In antiquity and in medieval times, logic and mathematics had been considered as two quite independent disciplines.¹⁴ Even though it is probable that Aristotle elaborated his theory of syllogisms drawing inspiration from contemporary ideas in mathematics, and even though Stoic arguments were employed by later commentators to justify certain theorems in Euclid's *Elements*, the idea of reducing logic to a calculus seems to have been conceived for the first time in the period between the second half of the sixteenth and the first half of the seventeenth century.¹⁵ Pierre de la Ramée (1515–72), for instance, claims that the Greek word for ‘syllogism’ properly means a calculus or a computation and Thomas Hobbes in his *Computatio sive Logica* plainly identifies the act of thinking with that of performing an addition or a subtraction: “By ratiocination, I mean computation. Now, to compute is either to collect the sum of many things that are added together, or to know what remains when one thing is taken out of another. Ratiocination, therefore, is the same with addition and subtraction.”¹⁶ Clearly, Hobbes's idea that logical operations simply amount to adding or subtracting concepts, was too naïve to permit any interesting development in the direction of a logical calculus. Yet, the claim that reasoning is the same as computing was quite new and invited people to create stronger relationships between logic, understood as the discipline which dictates the canon for correct reasoning, and mathematics.

and obscurity” which afflict the language of (scholastic) logicians. To these he opposes the clarity and simplicity to which mathematicians are accustomed. This contraposition is typical of the humanist thinkers and, as far as I know, there is no trace of it in logic handbooks written before the fifteenth century.

¹² The translation concerned the first six books only.

¹³ Pelletier (1557: 12).

¹⁴ Cf. Muller (1981); Acerbi (2010: 15–86).

¹⁵ On the relationships between logic and mathematics in the seventeenth century, see Mugnai (2010).

¹⁶ Hobbes (1839a: 3). We find an analogous claim concerning syllogism in Gassendi (1658, vol. 1: 34).

Another important consequence of re-introducing Euclid's works in Western culture was that together with these works even some ancient commentaries on the *Elements* (like Pappus's commentary, for instance; fourth century AD), were to become widely known. Relying on Pappus, the astronomer and mathematician Christopher Clavius (1538–1612) accordingly contributed to an increasing interest in the distinction between *analysis* and *synthesis* among philosophers and mathematicians.¹⁷

Roughly speaking, given a certain mathematical problem, *analysis* was usually described as a method for going backwards from the problem to some general principles that were involved in it and which could constitute a basis for finding a solution. *Synthesis* was the opposite: from the principles to the solution. *Analysis* was intended as a kind of heuristic process, whereas *synthesis* was devised as a method for proving theorems starting from first principles and axioms. Thus, analysis and synthesis were viewed as two essential tools for the project of renewing logic both as an art of judgment (*ars iudicandi*) and as an art of discovery (*ars inveniendi*).

The processes of analysis and synthesis played an essential role in another project characterizing the cultural climate at the beginning of the seventeenth century: the project of the creation of a universal language. This project, in turn, was linked to the old tradition of so-called *Lullism*, from the medieval thinker Ramon Lull (1232–1316), who is credited as a founding father of combinatory and as a source for Leibniz's ideas on combinatorial art.¹⁸

In his *Ultimate General Art* (*Ars Magna*, 1305), Ramon Lull proposes a program the main features of which may be summarized as follows. Once you have the first concepts, which he believes are nine in number and correspond to the properties of God (Goodness, Power, Wisdom, Truth, Glory, etc.), Lull proposes the construction of a machinery composed by concentric discs of paper rotating around a rigid axle. Each disc (properly a circle) is divided into nine sectors and in each sector a letter is typed representing a concept. The outer disc is fixed and contains the letters representing the first concepts from the combination of which all others are generated. The inner circles, which contain letters of other general concepts, different from those corresponding to the divine properties, may each revolve in opposite direction with respect to the others. Moving the inner circles and keeping the outer one fixed, the 'artist', i.e. the expert or

¹⁷ For the development of the concept of analysis in Western philosophy, see Beaney (2016).

¹⁸ On Lull's art and logic, see Bonner (2007).

master of the art, could create all possible combinations of the letters printed in the sectors. With two circles, he might give rise to all possible combinations of *two* letters; with three circles to all possible combinations of *three* letters, etc. Lull's idea was that using this device systematically, one could reproduce all complex concepts of all created things.¹⁹ Lull considered the *General Art* as a tool for producing *true knowledge* and believed that simply manipulating a few discs of paper and combining their inscribed letters one could discover new truths in every field of human knowledge. His aim was that of creating a general theory of the principles of all sciences: a kind of 'science of all sciences', the nature of which was not merely formal and which situated itself in a terrain intermediate between logic and metaphysics.²⁰

Starting from the second half of the sixteenth century, a renewed interest in Lullism was largely motivated by the reaction against traditional ways of teaching (Aristotelian) scholastic logic. The latter seemed too formal and too remote from real life to be of any interest, whereas the Lullian *Art*, which appeared to be playing with concepts of *real* properties and was hunting for truths concerning real beings, was more promising in the eyes of people who attempted to give a practical orientation to philosophy.

3 The project for the creation of a universal language

As we have seen, Lull employed letters of the Latin alphabet to represent concepts, but other signs or characters could also be invented to denote the 'primitive concepts'. One can, for instance, devise some set of particularly simple signs, easy to be manipulated and remembered, and then generate, according to certain rules, the complex configurations corresponding to all complex concepts. In this way, each concept could be associated with a kind of 'picture', explicitly representing the degree of complexity of the concept itself. This latter is a typical example of what, at the beginning of the seventeenth century was considered a project for the construction of a *universal language*. The connection with the *Art* of Ramon Lull was not a necessary requisite for the project: many authors, however, were driven to associate the two because of contingent analogies, mainly grounded on the

¹⁹ This is a quite simplified account of Lull's art: for a more accurate explanation, see Bonner (2007).

²⁰ Cf. Bonner (2007: 127) and Chapter 5 in particular.

recourse to the methods of analysis and synthesis. This is the case, for example, with Seth Ward (1617–89), Savilian Professor of astronomy at Oxford, who had an interest in Lull's theories and gave a very clear exposition of the project for a universal language:

[...] for all Discourses being resolved in sentences, those into words, words signifying either simple notions or being resolvable into simple notions, it is manifest, that if all the sorts of simple notions be found out, and have Symbols assigned to them, those will be extremely few in respect of the other, (which are indeed Characters of words, such as *Tullius Tiro's*)²¹ the reason of their composition easily known, and the most compounded ones at once will be comprehended, and yet will represent to the very eye all the elements of their composition, and so deliver the natures of things: and exact discourses may be made demonstratively without any other pains than is used in the operations of specious Analytics.²²

'Specious analytics' (*analytica speciosa*) is a name also employed by Leibniz in the DAC to designate the *algebra* introduced by François Viète (1540–1605); the term *speciosa*, a derivation of *species*, alluding to the possibility of dealing simultaneously with several things of the same kind. As Seth Ward stresses, the creation of a universal language requires first that we are able to decompose our complex concepts into their component parts. To execute this task, however, we need something like a register of all (or at least of the most important) complex concepts. Once this register was ready, it would contain the greatest part of human knowledge all at once. It is from this idea that the project for the construction of an encyclopedia took its origin, an encyclopedia that would contain the entire knowledge acquired by human beings in the course of the centuries. Thus, in the period considered, Lullism, the search for a universal language and the project for constructing an encyclopedia were three different, but fairly interconnected subjects, pursued by a considerable number of philosophers in Europe. Amongst these, Johann Heinrich Alsted (1588–1638), the celebrated author of a huge encyclopedia in seven volumes (Herborn 1630), and an author well-known to the young Leibniz, played a very influential role.²³

²¹ Marcus Tullius Tiro (first century BC) was first a slave and then a freeman of Cicero who invented the 'Tironian notes', the first Latin shorthand system.

²² Ward (1654: 21).

²³ On Alsted's influence on the young Leibniz, see Loemker (1961); Mugnai (1973). For an exhaustive account of the relationships between Alsted and Leibniz on matters related to

As Alsted writes in his *Clavis Artis Lullianae et verae logicae* (*A Key for the Lullian Art and the True Logic*, 1609), Lullism was poorly represented in Germany at the time, whereas it was flourishing in Spain, France, and Italy. In Italy, in particular, many philosophers (here called *speculatores*, ‘speculators’) applied themselves intensively to the study of this *art*, becoming a kind of addict to it in the course of doing so.²⁴ Alsted calls Ramon Lull “a mathematician and a cabbalist”²⁵ and praises him for the use of letters of the alphabet to denote concepts, as well as for his recourse to mathematical symbols like circles. He considers the former as valid aids to memory and the latter as tools on which the teaching of any science is based.²⁶

In comparison with other texts devoted to the same topic (Bruno’s works included), Alsted’s *Clavis* constitutes a rare example of clarity and conciseness: it gives a terse account of how Lull’s art works and weds the Lullian combinatorial with some features borrowed from the reformed logic proposed by Pierre de la Ramée (1515–72), thereby giving rise to a logical system mainly oriented at the development of topical arguments, i.e. arguments conceived with the purpose of ruling dialectical disputes.²⁷

4 Atomism and ‘immeation’

Compared with the works of authors belonging to the tradition of Lullism, the *DAC* differentiates itself first of all because of the emphasis put on the theory of syllogism, then for the attempt to extend the combinatorial calculus to any kind of matter. Finally, even though Leibniz’s mathematical training was not very developed at the time, an intense mathematical flavour that can hardly be found among the followers of Ramon Lull emanates from the *DAC*.

According to the image of the world that we get reading the *DAC*, everything having a complex structure is a whole which can be decomposed into lesser wholes and these latter, in turn, into their least components parts

religion, see Antognazza-Hotson (1999); on Alsted’s encyclopaedism and the ‘second Reformation’ in Germany, see Hotson (2000). On Lullism, the encyclopedic projects and the universal language, see Rossi (2006).

²⁴ Alsted (1609: 19).

²⁵ Alsted (1609: 20).

²⁶ Alsted (1609: 21). For a further development of the doctrine, the reader of the *Clavis* is invited to consult the commentaries to the *Art* written by Giordano Bruno.

²⁷ On the relationships between Ramus’s doctrines and the Herborn encyclopedists (on Ramus’s philosophy in general), see Ong (2004).

(atoms or some molecular aggregates of some sort). By a simple process of the combination of the parts, one may, in turn, re-compose the whole. The same procedure can be applied to non-material things as well: to concepts, for example, propositions, geometrical figures, and numbers. Thus, combinatory applied to mereology, the doctrine of wholes and parts, becomes a kind of all-purpose tool in Leibniz's hands; a tool that allows one to reveal the intimate secrets of nature:

Since all things that exist, or can be thought, may be said to be made up of parts, either real or at least conceptual, whatever differs in kind must necessarily differ either in parts, and here lie the Applications of *Complexions*, or by a different situs, hence the application of *Dispositions*.²⁸

The system of variations elaborated in the *DAC*, Leibniz observes, "leads the mind that yields to it almost through all infinity, and embraces at once the harmony of the world, the inner workings of things, and the series of forms."²⁹ Introducing the seventh specimen, devoted to the application of combinatory to geometrical figures, Leibniz emphatically writes:

With these complications, not only can geometry be enriched by an infinite number of new Theorems, as every complication brings into being a new, compound figure, by the contemplation of whose properties we may devise new theorems and new demonstrations; but (if it is indeed true that great things are made up of little things, whether you call them atoms or molecules) we also have a unique way of penetrating into the arcana of nature. This is because the more one has perceived of the parts of a thing, the parts of its parts, and their shapes and arrangements, the more perfectly one can be said to know the thing. The structure of these figures is first to be investigated abstractly, in plane and solid geometry: after which, when you enter upon natural history and the question of being, that is, upon the question of the real constitution of bodies, the vast portals of Physics will stand open, and the character of the elements, the origin and mixture of the qualities, the origin of mixtures, the mixing of those mixtures, and everything that formerly lay hidden in darkness will be revealed.³⁰

²⁸ *DAC*: 93 (A VI, 1: 177).

²⁹ *DAC*: 133 (A VI, 1: 187).

³⁰ *DAC*: 135 (A VI, 1: 187).

Later in life, Leibniz would come back to the fact that, for a brief period in his youth, he had favoured atomism. That he had done so, is quite in agreement with the parenthetical remark made in the passage above, which shows no prejudice against atoms as the least components of natural bodies.³¹ Leibniz's atomism, however, is wedded in the *DAC* with a metaphysical doctrine centered on the notion of *immediation* or *perichoresis*, according to which everything is related with everything on the basis of the two relations of *similarity* and *dissimilarity*. 'Perichoresis' is the transliteration into English of a Greek word meaning 'circle' or the effect of revolving, which was employed in theology "to explain the relation of 'coherence' of the three persons of the Trinity".³² The Latin word 'immediatio' was coined to translate 'perichoresis' and according to Johann Heinrich Bisterfeld, Leibniz's main source on this issue, it was meant to designate "the varied concourse, combination and complication of relations".³³ In the *DAC*, Leibniz praises Bisterfeld's work:

I shall finally and briefly summon everything that can be culled from the metaphysical doctrine of the relations of Being to Being in order to form Places out of the different kinds of relation, and appropriate maxims characteristic of each of them, out of the theorems. I believe that I have seen this done, in more than just the summary fashion of encyclopaedists, by Johann Heinrich Bisterfeld in his *Universal Lamp, or Epitome of the Art of Meditating*, published in Leiden, 1657, which is entirely based on what he calls *immediation* and *perichoresis*, that is to say, the universal likeness of all things in all things, as well as the universal unlikeness of all things to all things, the principles of which are Relations. Anyone who reads that book will gain an increasing understanding of how to apply the art of complication.³⁴

Thus, Leibniz deeply modifies the account of the world usually associated with atomistic doctrine, which was based on the aggregation and disaggregation of atoms governed by mere chance, without any trace of intrinsic

³¹ On Leibniz's atomism, see Garber (2009: 62–70, 81–2); Arthur (2016).

³² Antognazza (1999: 50).

³³ Bisterfeld (1657a; 185–6) (Antognazza's translation; Cf. Antognazza (1999: 50)). As Maria Rosa Antognazza remarks, Bisterfeld, in his recognition of the universal analogy with Trinity, "was certainly not alone. Rather he was working within a long tradition rooted in Augustinian thought." Ideas similar to those of Bisterfeld are to be found in Alsted and in "that other and still more illustrious student of Alsted: Jan Amos Comenius." (Antognazza 1999: 52).

³⁴ *DAC*: 70.

finalism. To this view he substitutes the picture of a world composed of beings connected by a net of reciprocal *relations* that generate a mutual union and communion analogous to that of the three persons in the holy Trinity. This picture was clearly meant to avoid the materialistic and atheistic consequences implicit in genuine atomistic theories as, for example, the view displayed by Lucretius in his poem *On the nature of things*. It is not difficult to see the existence of a latent conflict between Leibniz's acceptance of atomism on the one hand, and his agreement to a metaphysics motivated by religious issues and influenced by a philosophy largely inspired by doctrines of Neo-platonic origins on the other.³⁵ As soon as the conflict came to light, in the years immediately following the edition of the *DAC*, Leibniz decided to abandon the atomistic theory, but in a certain sense he never stopped flirting with it: when he attempts to explain to his contemporaries his *hypothesis of the monads*, he uses the expression 'formal atom' to characterize what he properly intends to denote with the word 'monad'.³⁶ As formerly in the *DAC*, in his mature thought Leibniz attempts to reconcile an atomistic perspective with an anti-materialistic metaphysics.

5 Themes from the DAC in Leibniz's mature philosophy

Besides the presence of some ideas that will constitute a substantive part of the project for the *characteristic art* later on, and besides an inclination to mix atomism with a strongly anti-materialistic perspective, we find in the *DAC* other items clearly recognizable in Leibniz's mature works as well. Among them, the following are particularly worth mentioning:

- 1) The notion of *blind thought*;
- 2) The distinction between propositions that are eternally true and propositions the truth of which is founded "not on essence but on existence";
- 3) A detailed and systematic treatment of the theory of the syllogism.

Leibniz calls 'blind thought' (*cogitatio caeca*) any thought that is developed essentially by means of words or other symbols. Such a thought is *blind*

³⁵ On the influence of Neoplatonic doctrines on Leibniz's thought, see Mercer (2000).

³⁶ Cf., for instance, what Leibniz writes in the *New System of the Nature of Substances and Their Communication* (1695): "So, in order to get to these real unities I had to have recourse to a formal atom, what might be called a real and animated point, or to an atom of substance, which must contain some kind of form or activity in order to make a complete being" (*Philosophical Texts*: 143).

because by employing words or any other kind of symbols, we *do not see* our own thoughts, which remain concealed behind a curtain of material signs, spoken or written or whatever.

The category of *blind thought* plays a prominent role in Leibniz's mature philosophy. Even early on during his stay in Paris, Leibniz emphasizes the fact that human beings cannot grasp directly by an act of intuition any thought beyond a certain degree of complexity.³⁷ In other words, we are forced to take recourse to symbols to formulate and to express concepts or thoughts sufficiently complex. We need a language, or at least something like a language, not only to communicate, but even to develop our own thoughts.³⁸

The idea that some kind of material tool is necessary for thinking is pervasive in Leibniz's philosophy and strictly connected with the opinion according to which, with the only exception of God, all spiritual beings (even the angels) are associated with a body (even though a very 'thin' one). According to the Platonic tradition, Leibniz thinks that this 'bodily part' prevents any kind of spiritual beings from being directly acquainted with ideas and thoughts in their 'purity', as it were.

The distinction between propositions that are eternally true and propositions the truth of which is founded "not on essence but on existence" foreshadows the distinction between 'truths of reason' and 'truths of fact' as it is expressed, for example, in the *New Essays* (1702).³⁹ In the *DAC*, the propositions "that are eternally true" are synonymous with "theorems", i.e. "propositions that hold not because of the will of God, but by their very nature", and are contrasted with "singular propositions" like "Augustus was emperor of the Romans" and with "observations", i.e. "universal propositions whose truth is founded not on essence but on existence".⁴⁰ In a slightly different form, we find the same distinction in the so-called *Monadology*:

There are two kinds of truths, those of *reasoning* and those of fact. Truths of reasoning are necessary and their opposite is impossible, while those of fact are contingent and their opposite is possible. When a truth is necessary one can find its reason through analysis, resolving it into ever more simple ideas and truths until one reaches primitives.⁴¹

³⁷ Cf., for instance, A VI, 3: 462.

³⁸ This does not exclude the possibility that, in certain peculiar cases, we may have direct acquaintance with some ideas or truths, see Mugnai (2014).

³⁹ Cf., for instance, NE: 446.

⁴⁰ *DAC*: 177 (A VI, 1: 199).

⁴¹ *Monadology*, section 33 (transl. Rescher 1991: 120).

In the *New Essays*, Leibniz explicitly considers the distinction between propositions of reason and propositions of fact as 'his own',⁴² and in the same vein as in the *DAC*, he writes that propositions of fact "can also become general, in a way; but that is by induction or observation, so that what we have is only a multitude of similar facts [...]". General propositions of reason, instead, "are necessary".⁴³

Finally, the *DAC* emphasizes Leibniz's precocious interest in the theory of syllogism. Even though Leibniz in 1686 had worked out a logical calculus, which was supposed to include the traditional syllogistic theory, he never stopped investigating the logical mechanisms governing the syllogistic doctrine of Aristotelian origin. It is quite remarkable that late in his life, only one year before his death, Leibniz devoted himself to writing an essay on the syllogistic figures, aiming to give a new rigorous foundation to traditional syllogistic theory.⁴⁴ In the *New Essays* we find an apology for the syllogism:

It must be admitted that the scholastic syllogistic form is not much employed in the world, and that if anyone tried to use it seriously the result would be prolixity and confusion. And yet – would you believe it – I hold that the invention of the syllogistic form is one of the finest, and indeed one of the most important, to have been made by the human mind. It is a kind of universal mathematics whose importance is too little known. It can be said to include an *art of infallibility*, provided that one knows how to use it and gets the chance to do so – which sometimes one does not.⁴⁵

Many years before the *New Essays*, when he began to write the first texts on logical calculus, Leibniz elaborated a very general notion of *logically valid* arguments, which includes the traditional syllogisms as a particular case. This point, too, is clearly stated in the *New Essays*:

But it must be grasped that by 'formal arguments' I mean not only the scholastic manner of arguing, which they use in the colleges, but also any

⁴² "The distinction you draw appears to amount to mine, between 'propositions of fact' and 'propositions of reason'." (*NE*: 446).

⁴³ *NE*: 446.

⁴⁴ This essay has recently been published by Wolfgang Lenzen in *Schriften* (2019: 428–43). A partial edition of it appeared in the *Leibniz Review*, vol. 20, 2010: 117–35.

⁴⁵ *NE*: 478.

reasoning in which the conclusion is reached by virtue of the form, with no need for anything to be added. So: a sorites, some other sequence of syllogisms in which repetition is avoided, even a well drawn-up statement of accounts, an algebraic calculation, an infinitesimal analysis – I shall count all of these as formal arguments, more or less, because in each of them the form of reasoning has been demonstrated in advance so that one is sure of not going wrong with it.⁴⁶

We have thus, with a ‘broad brush’, given a picture of the cultural background against which Leibniz conceived the *DAC*; it is time now to enter into some details concerning the main topics discussed in this work.

6 The first two problems and their solutions

Leibniz begins the *Dissertation* with the clear statement that his investigation properly belongs to *metaphysics*.⁴⁷ Metaphysics, indeed, “is concerned first with Being, and then with the affections of Being”;⁴⁸ and it is on these latter that the doctrine of combinations rests. Leibniz does not say explicitly what he means by the words ‘being’ (Latin: *ens*) and ‘affection’ (*affectio*, *modus*), but tacitly assumes some familiarity on the part of the reader with these terms and writes that just as affections of bodies (colour, weight, and figure) are not themselves bodies, so the affections of a being are not beings.

According to the scholastic doctrine (inspired by Aristotle’s *Categories*), Leibniz considers the totality of the affections as divided into three main categories: *quality*, *quantity*, and *relation*. A *quality* is an “absolute affection (or mode)” of a being; a *quantity* is a relative affection “which holds between a thing and one of its parts”; a *relation* is an affection, which holds between a thing and some other thing. If metaphysics is understood as the discipline that investigates “what is common to all kind of Beings”, then *quality*,

⁴⁶ *Ibidem*.

⁴⁷ As Eberhard Knobloch observes: “Leibniz [...] spoke of the “ars combinatoria”, a philosophical term to which he attached different meanings in the course of his life, but which always embraced more than what we call combinatorics today and was not even restricted to mathematical problems.” (Knobloch 1974: 410)

⁴⁸ *DAC*: 69 (A VI, 1: 170).

quantity, and *relation*, because they concern the most general affections of all things, belong to metaphysics as well.

After introducing these three categories, Leibniz recognizes two kinds of relations: *unity* and *agreement*. We have a relation of *unity*, when several things are put together to form a *whole*: in this case, the things forming the whole are called *parts*. Leibniz seems to have quite a general idea of a whole, as something that obtains by simply juxtaposing or adding things together: it does not necessarily presuppose some kind of intrinsic link among the parts.⁴⁹ Moreover, Leibniz does not limit himself to considering *material* wholes only: on the contrary, he admits any kind of wholes, even *immaterial ones*, such as a piece of music or a poetical work. He is quite reticent about the relation of *agreement*, probably because it does not play an important role in the art of combinations. In later texts, written after the edition of the DAC, the relation of *agreement* (Lat. *convenientia*) seems to be not much different from *similarity under some respect*. Thus, for example, two ‘things’ agree if they have properties (colour or shape) that are similar or if they share a similar structure.⁵⁰

Given a whole, it can be divided into lesser parts and these parts can be rearranged in different ways. Thus, for instance, let a whole consist of A, B, and C. The lesser wholes “are its parts AB, BC, and AC; and the disposition of the smallest parts, or what are taken as the smallest parts (that is, Unities), can be varied both in relation to each other and to the whole”.⁵¹ Leibniz calls *variation* the change of relations (including change of order) amongst the parts of a given whole (what we now call *permutation*); and *complexion*, the “union of a lesser whole within a greater whole” (what we now call *combination*). Loosely speaking, a *complexion* is the grouping of the elements of some domain of things (abstract or material), according to a given number: thus, for example, given a domain of four elements, we have six *complexions* of two elements out of them. To indicate that a domain (a whole) made of x elements must be decomposed into complexions each of n elements, Leibniz uses the notation ‘ x^n ’ (where ‘ x ’ is the number of the things belonging to the intended domain). The disposition of the parts in relation to each other and to the whole is called *situs*.

⁴⁹ Around 1686, Leibniz will assume that a part should be ‘homogeneous’ to its whole; see, for instance, A VI, 4B: 1671 (discussion with Fardella (1690)): “because a part is always homogeneous with the whole”.

⁵⁰ Cf. A VI, 4A: 336: “A relation of agreement is either of similarity or dissimilarity.”

⁵¹ DAC: 73 (A VI, 1: 171).

After introducing the main notions on which the combinatorial art should rest, Leibniz proposes a list of problems preceded by a short pre-ambles, which explains how he intends to develop his investigation:

There are three things that have to be considered: *Problems*, *Theorems*, and *Applications*. To some of the problems I have appended an application and, where it seemed worthwhile, theorems. Moreover, to some of the problems I have added an explanation of the solution.⁵²

Thus, Leibniz considers a totality of twelve problems and a variety of applications, ranging from the theory of syllogism to some puzzles typical of recreational mathematics as, for instance, how many ways a given number of persons could sit down at one table in some order or other. Problems 1–3 concern combinations, whereas problems 4–12 are centred on permutations. The chapters are of different length and some of them quite disproportionate, when compared to others. As Knobloch remarks:

The longest chapter (problem 2) alone takes up 28 pages of the Academy edition, while the shortest fills a mere 2 1/2 lines. On the other hand, the final paragraph following the formulation of the 12th problem refers to all of the last 6 problems, whereas the second chapter includes part of the first.⁵³

Among the applications, the most interesting ones from a philosophical point of view are those concerning logic and the syllogism, and it is on these that we will concentrate in these introductory remarks.

The first problem (PROBLEM I) tackled by Leibniz is:⁵⁴

(I) *given a number with its corresponding exponent, to find the complexions.*

Leibniz offers two solutions: one concerning the general case, when the exponent is any number whatsoever, the other involving the particular case with exponent 2. The first solution sounds quite cumbersome:

⁵² Cf. DAC: 79 (A VI, 1: 173).

⁵³ Knobloch (1974: 412).

⁵⁴ To distinguish the main problems into which the DAC is articulated from the problems internal to the various applications, we write in capitals the first.

Let the number of complexions of the preceding exponent with the preceding number be added to the number of complexions of the given exponent with the preceding number; the result will be the required number of complexions.⁵⁵

To understand Leibniz's words, let us put the number of *complexions* = 'C'; the number of 'things' to be combined = '*n*'; and the number of the exponent = *k*. In this case, '*n^k*' denotes all complexions of *k* elements out of *n* things. If we suppose *n* = 5 and *k* = 3, then '5³' denotes all complexions (combinations) of 3 elements that can be made out of 5 elements. Leibniz's proposed solution corresponds to the formula:

$$(F1) \ C\left(\begin{matrix} n \\ k \end{matrix}\right) = C\left(\begin{matrix} n-1 \\ k-1 \end{matrix}\right) + C\left(\begin{matrix} n-1 \\ k \end{matrix}\right)$$

As this formula shows, the total number of complexions of *k* elements out of *n* 'things' is determined by adding the number of complexions of *k*–1 elements out of *n*–1 things to the number of complexions of *n* elements out of *n*–1 things. To stick to our example, to calculate the number of complexions of three elements out of five things, we must: a) calculate the number of complexions of two elements out of four things; b) calculate the number of complexions of three elements out of four things; and, finally; c) add together these two numbers. In this case, however, to determine the number of complexions of *k* elements out of *n* things, we need to know the number of complexions of *n*–1 things: as Leibniz remarks, to find the complexions, "the complexions of the preceding number are a prerequisite".⁵⁶

After having expressed in words the content of (F1), Leibniz proceeds to construct a table devised as a tool to find all possible complexions of *k* elements out of *n* 'things', for *k* and *n* comprised between the numbers 1 and 12.⁵⁷ It is to this table that Leibniz constantly refers, when he proposes the solutions to the combinatorial problems formulated in the DAC.

⁵⁵ Cf. DAC: 81 (A VI, 1: 174).

⁵⁶ DAC: 83 (A VI, 1: 174). The total number of complexions can be found in a more general and direct way applying the formula (unknown to Leibniz at the time of the DAC):

$$C\left(\begin{matrix} n \\ k \end{matrix}\right) = \frac{n!}{k!(n-k)!}.$$

⁵⁷ As a matter of fact, however, the column with the number of 'things' starts with '0', even though, as Knobloch (1974: 413) remarks, Leibniz "explicitly excludes combinations of no elements since he gives the number of all possible combinations as 2^{*n*}–1".